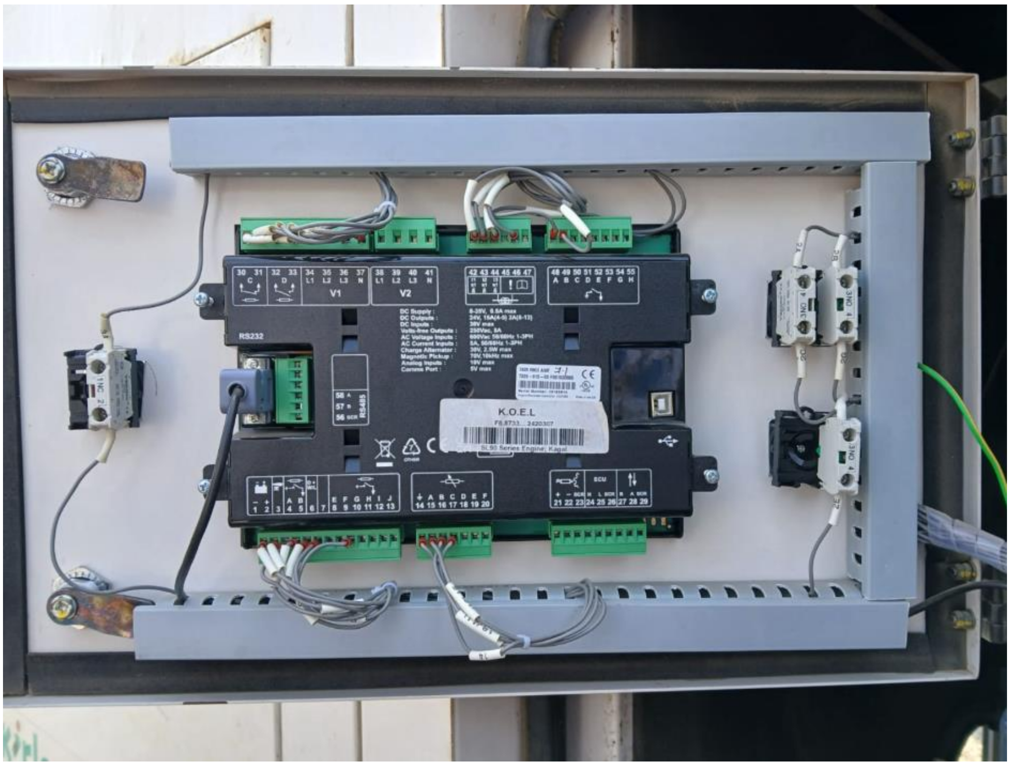
SITE STUDY

1. Objective:

To investigate the communication capabilities of the Kirloskar DG controller and determine the feasibility of acquiring operational data for remote monitoring.

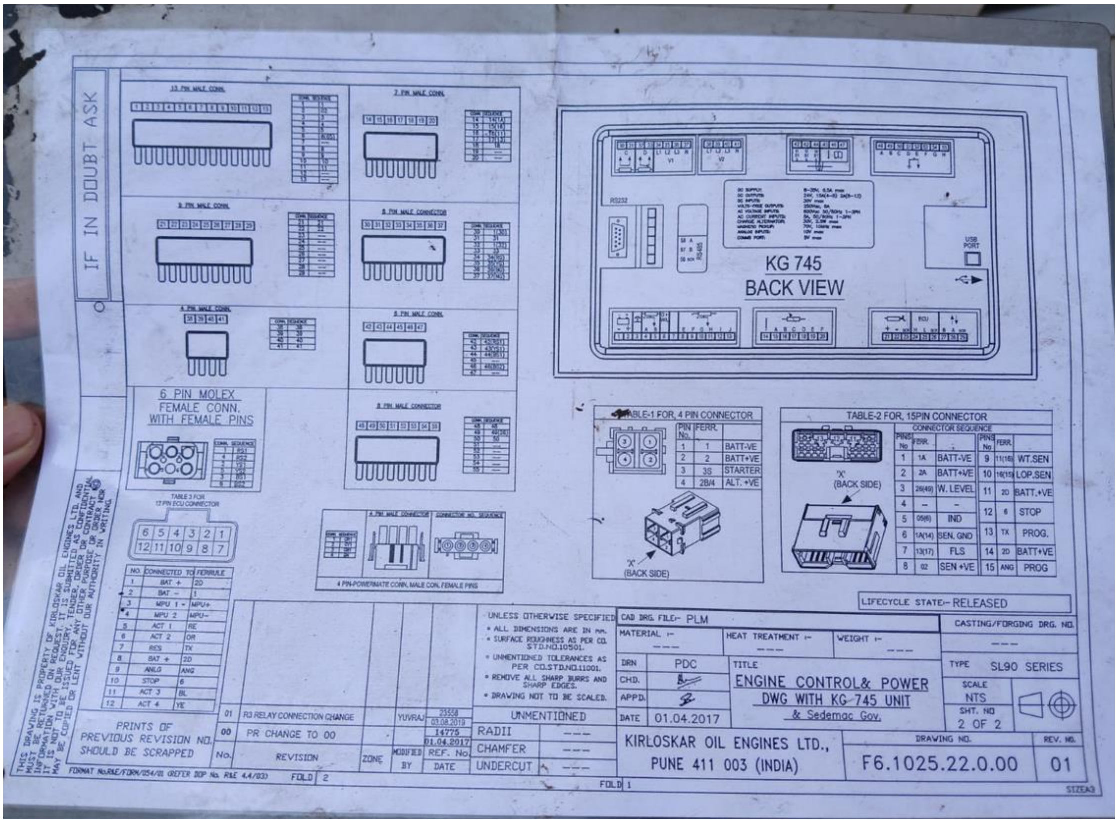
2. Generator and Controller Details:

Generator Manufacturer: Kirloskar
Controller Model: KG745



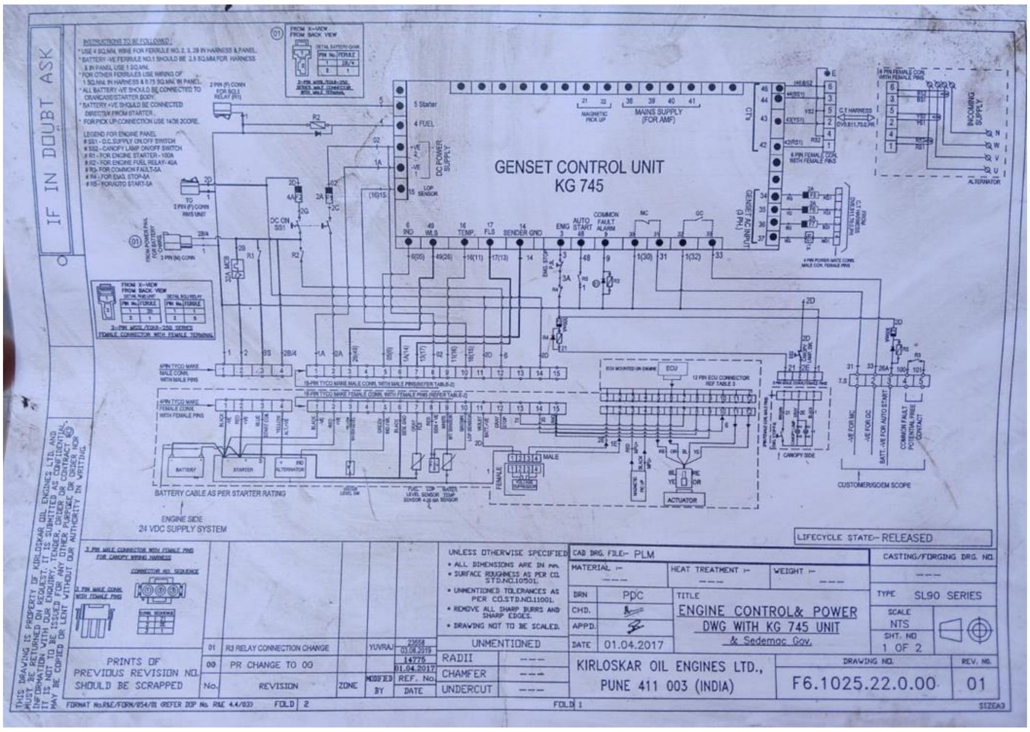
3. Initial Observations

• DG set was operational.
• Controller model identified as KG645.
• Communication port available on controller.
• Controller supports Modbus RTU communication.
• RS485 interface identified for data communication.



4. Communication Interface Study

Parameter	Observation
Communication Protocol	Modbus RTU
Physical Layer	RS485
Connector Type	Terminal Block
Slave ID	1
Baud Rate	9600
Parity	None



5. Data Availability Assessment

Parameters identified as available through controller:

Fuel Level, Engine RPM, Battery Voltage, Coolant Temperature, Generator Voltage, Generator Current, Power Output, Running Hours, Alarm Status

6. Proposed Solution

DG Controller

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RS485 Modbus RTU

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MAX485 Converter

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ESP8266

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MQTT

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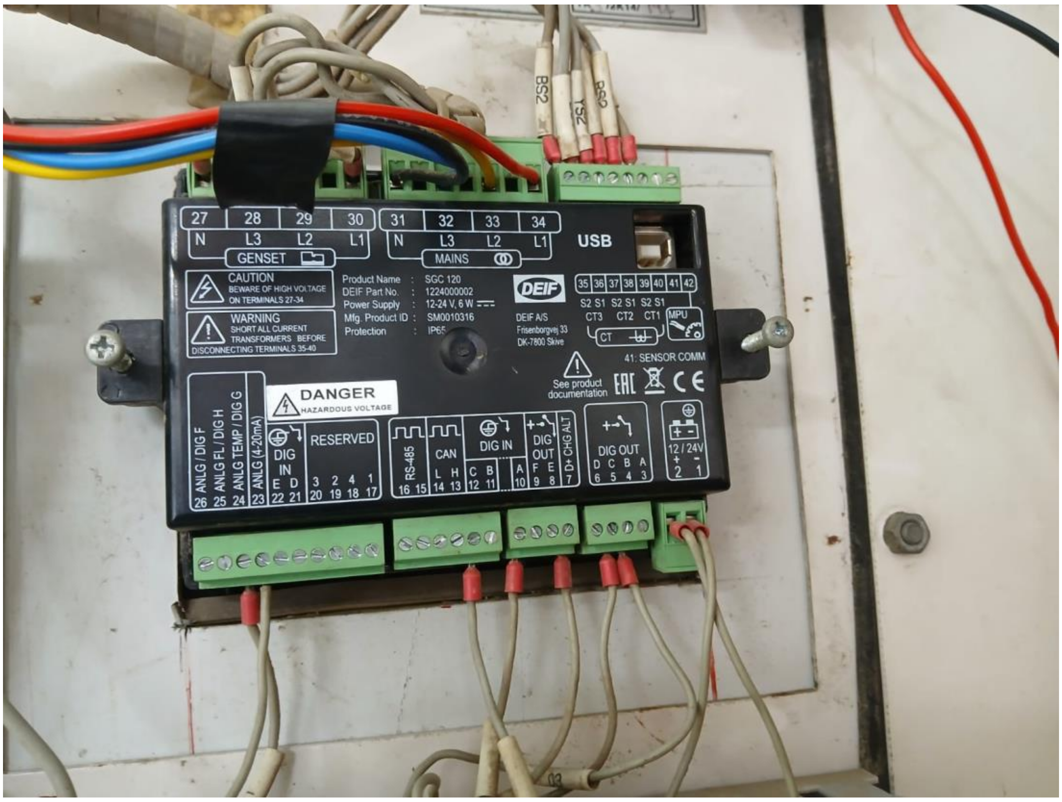
Monitoring Dashboard

SITE PHOTO


Accurate Powertech India Pvt. Ltd.


BS2
YS2
RS2
27 28 29 30
N L3 L2 L1
GENSET
31 32 33 34
N L3 L2 L1
MAINS
USB
CAUTION
BEWARE OF HIGH VOLTAGE
ON TERMINALS 27-34
WARNING
SHORT ALL CURRENT
TRANSFORMERS BEFORE
DISCONNECTING TERMINALS 35-40
Product Name : SGC 120
DEIF Part No. : 1224000002
Power Supply : 12-24 V, 6 W
Mfg. Product ID : SM0010316
Protection : IP65
DEIF
DEIF A/S
Frisenborgvej 33
DK-7800 Skive
35 36 37 38 39 40 41 42
S2 S1 S2 S1 S2 S1
CT3 CT2 CT1
CT
MPU
41: SENSOR COMM
See product documentation
DANGER
HAZARDOUS VOLTAGE
ANLG / DIG F
ANLG FL / DIG H
ANLG TEMP / DIG G
ANLG (4-20mA)
DIG IN
E D
RESERVED
3 2 4 1
26 25 24 23 22 21 20 19 18 17
RS-485
CAN
L H
DIG IN
C B A
DIG OUT
F E
D+ CHG ALT
16 15 14 13 12 11 10 9 8 7
DIG OUT
D C B A
6 5 4 3
12 / 24V
+ -
2 1